Lecture 9 - Research Ethics of Causal Inference

Assessing Assumptions

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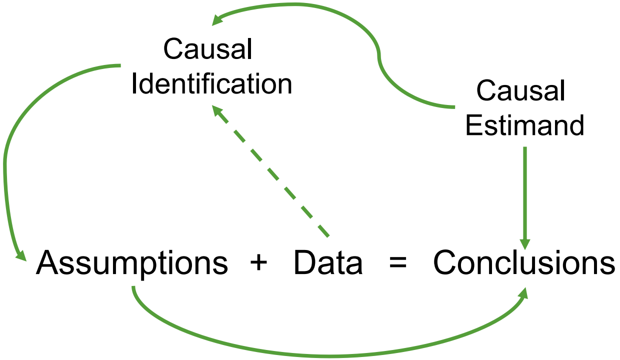
Outline

The role of assumptions in causal inference



- Assumptions are always needed.
- Different assumptions ⇒ Different conclusions.
- Credibility of assumptions ⇒ Credibility conclusions.

Assumptions determine conclusions

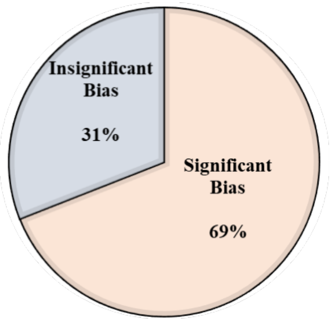
A recent study[1]:

- Asks teams of researchers to analyze the same data set - **football games**.
- The data include info. such as player clubs, birthday, height, weight, and positions...
- **Research question**: *are referees more likely to give red cards to dark- than light-skin-toned players*?

[1] Silberzahn, R. et al. (2018). Many analysts, one data set: Making transparent how variations in analytic choices affect results. Advances in Methods and Practices in Psychological Science, 1(3), 337-356.

Assumptions determine conclusions

Results of the study:



- Researchers make their own assumptions and get different results.
- "*What theoretical and/or statistical rationale was used for your choice of covariates?*"

Assumptions determine conclusions

Similar studies in other fields show consistent observations:

- Biology:
  - Gould, E. et al. (2023). Same data, different analysts: variation in effect sizes due to analytical decisions in ecology and evolutionary biology.
- Economics:
  - Breznau, N. et al. (2022). Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty. Proceedings of the National Academy of Sciences, 119(44), e2203150119.
- Neuroscience:
  - Botvinik-Nezer, R. et al. (2020). Variability in the analysis of a single neuroimaging dataset by many teams. Nature, 582(7810), 84-88.

1. Understanding assumptions in causal inference
   - Assumptions on potential outcomes are untestable
   - Assumptions can be subjective
   - The law of decreasing credibility
2. Assessing assumptions
3. Conclusions

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Examples of assumptions

Stable Unit Treatment Value Assumption (SUTVA)

For a completely randomized experiment (CRE), we must have:

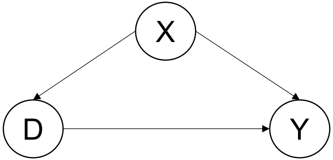
1. The potential outcomes for any unit do not vary with the treatments of others.
2. No different forms or versions of each treatment level.

Question: suppose we have a CRE, can we statistically test SUTVA with data?

Examples of assumptions

(Conditional) Unconfoundedness

The potential outcomes are independent from the treatment status (given a set of control variables X).

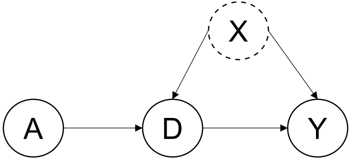


Question: With data on $\{Y, D, X\}$, can we statistically test the unconfoundedness assumption?

Examples of assumptions

Non-compliance (instrumental variables)

To identify the local average treatment effect, the treatment assignment (instrumental variables) must be 1) exogenous, 2) relevant, 3) excluded, and 4) monotonic (no defiers).

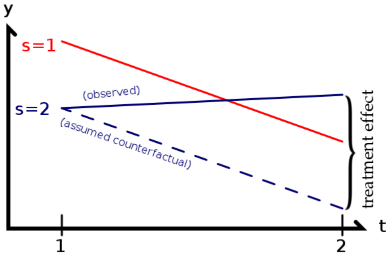


Question: With data on $\{Y, A, D\}$, can we statistically test the assumptions on instrumental variables?

Examples of assumptions

Parallel Trend Assumption in DID

In the absence of treatment, the difference between the treatment and control group is constant over time.

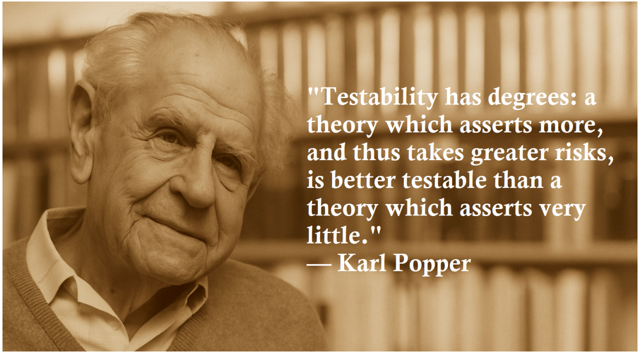


Question: With data on $\{Y_0, Y_1, D_0, D_1\}$, can we statistically test the parallel trend assumption?

Assumptions on potential outcomes are untestable!

Untestability of Assumptions

Given the fundamental problem of causal inference, any assumptions on potential outcomes are always statistically untestable.



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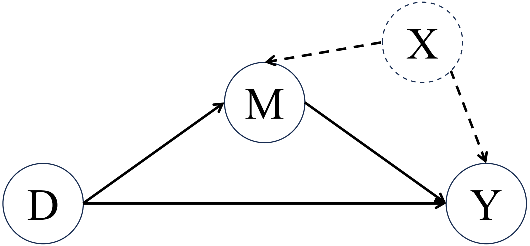
Assumpptions are not unique

The core problem of the traditional mediation analysis:

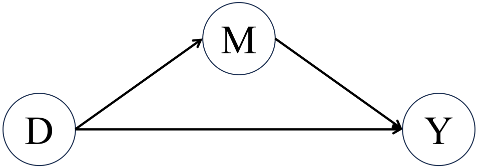


$$
\begin{cases}
Y &= \alpha_1 + \beta_1 D + e_1 \\
M &= \alpha_2 + \beta_2 D + e_2 \\
Y &= \alpha_3 + \beta_3 D + \beta_4 M + e_3
\end{cases}
$$

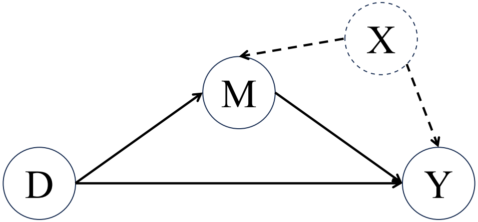
Question: what assumptions needed for the Baron and Kenny (1986)'s approach?

Assumpptions are not unique

(a) Baron DAG



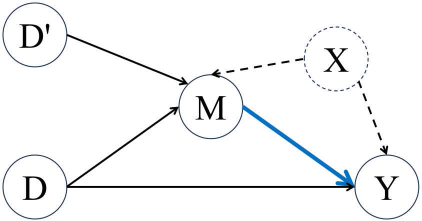
(b) Mediation DAG



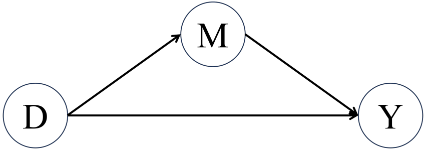
1. Treatment D is unconfounded.
2. Mediator M is unconfounded.

Assumpptions are not unique

(a) Constructed IV



(b) Baron DAG



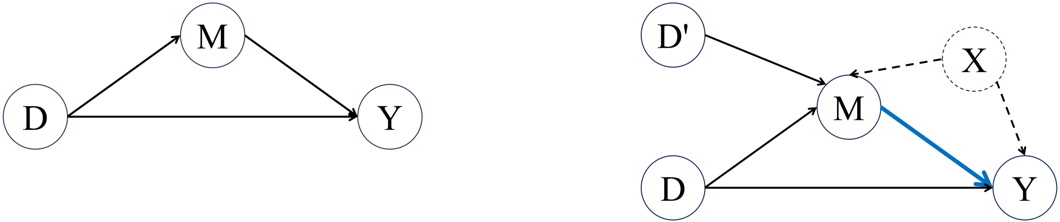
Constructed IV or D':

1. U is homoskedastic w.r.t. D or $\text{Cov}(De_2, e_3) = 0$.
2. V_2 is heteroskedastic w.r.t. D or $\text{Cov}(De_2, e_2) \neq 0$.

Assumpptions are not unique

(a) Baron DAG

(b) Constructed IV



Traditional Mediation	Constructed IV
$\rho\,(e_2, e_3) = 0$	$\rho\,(e_2, e_3) \neq 0$
$-11.7\ \ (-20.2, -4.8)$	$-5.2\ \ (-12.1, 0.4)$

A short discussion

Is a difference-in-difference design more credible than a simple regression?

A simple regression

$$Y_i = \beta D_i + \varepsilon$$

A DID regression

$$Y_{it} = \alpha_i + \lambda_t + \beta D_i T_t + \varepsilon$$

Where Y is the outcome, D is the treatment status, T is the before-after dummy.

The selection of assumptions is subjective

In Silberzahn et al. (2018), researchers reported the rationale of their analytical decisions:

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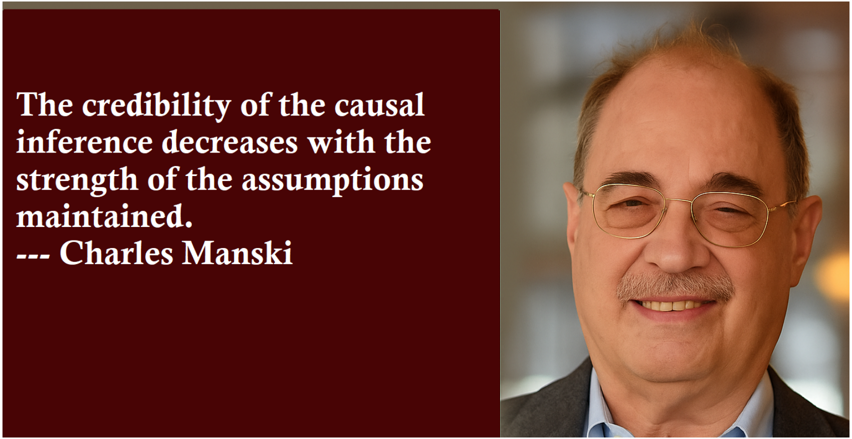
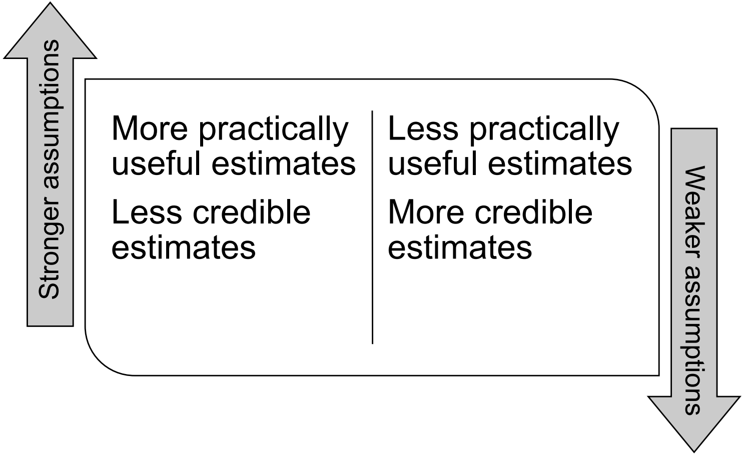
The law of decreasing credibility

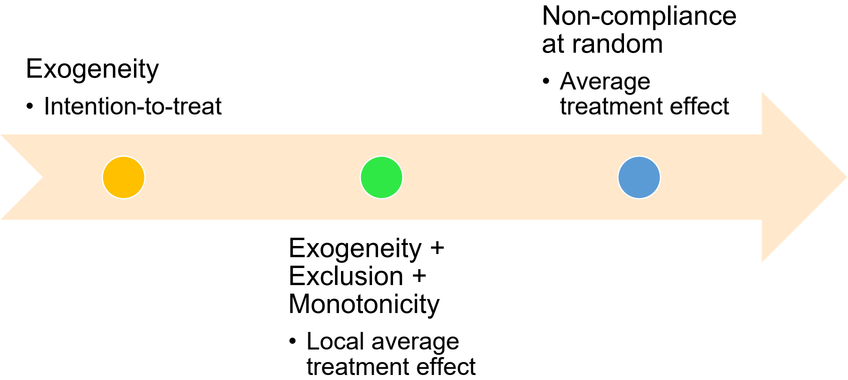


The law of decreasing credibility



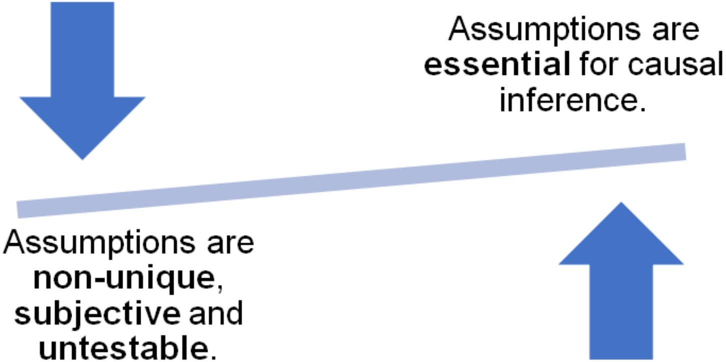
The law of decreasing credibility

Example (Non-compliance)

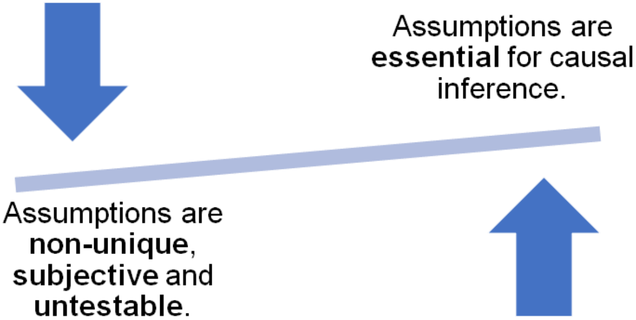


A recap on assumptions

We always need assumptions for causal inference practices!



A recap on assumptions



- How do we know the credibility of assumptions?
- What tools are available?

Tools in our toolkit

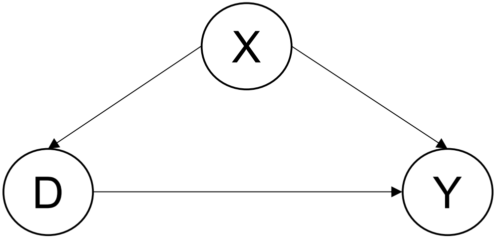
- The assumptions on potential outcomes are untestable.
- However, we can observe the implications of the assumptions.
- We assess assumptions by testing the implications.

Sensitivity analysis

Consistency test

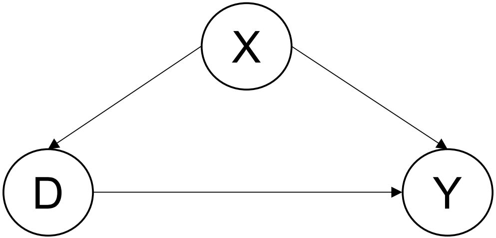
The basic idea of sensitivity analysis

- In practice, we often apply conditioning strategy and assume conditional unconfoundedness.



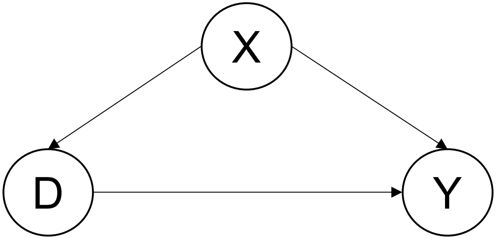
The basic idea of sensitivity analysis

- In practice, we often apply conditioning strategy and assume conditional unconfoundedness.
- This is a strong assumption, as there is always possible that confounders exist.



The basic idea of sensitivity analysis

- In practice, we often apply conditioning strategy and assume conditional unconfoundedness.
- This is a strong assumption, as there is always possible that confounders exist.
- **Question: if confounders exist, how they change the ATE estimation?**



The logic of sensitivity analysis

Assume a confounder exists as U

Simulate the values of $U \sim f(U \mid D, Y, X)$

Use both U and X and re-estimate $P(Y \mid D)$

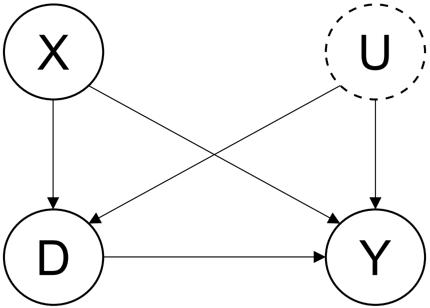
Examine the change in $P(Y \mid D)$ with vs. without U

The basic logic: the appeal to absurdity (*reductio ad absurdum*).

Sensitivity analysis: A toy model

A simple model with treatment D, outcome Y, observable X and a single unobservable $U \perp X$.

$$D := a_x X + a_u U$$
$$Y := b_x X + b_u U + \underbrace{\tau}_{\text{Objective}} D$$



Sensitivity analysis: A toy model

$$\begin{cases} D & := \alpha_x X + \alpha_u U \\ Y & := \beta_x X + \beta_u U + \tau D \end{cases}$$

If we observe both X and U, we can recover τ:

$$E\left(Y_1 - Y_0\right) = E_{X,U}\left[E\left(Y \mid D = 1, X, U\right) - E\left(Y \mid D = 0, X, U\right)\right] = \tau$$

Yet, we do not observe U. If condition on X only, we get:

$$E_X\left[E\left(Y \mid D = 1, X\right) - E\left(Y \mid D = 0, X\right)\right] = \tau + \underbrace{\frac{\beta_u}{\alpha_u}}_{\text{Bias}}$$

Sensitivity analysis: a toy model

Technical proof:

$$
\begin{aligned}
E_X\left[E\left(Y \mid D=d, X\right)\right] &= E_X\left[E\left(\beta_x X + \beta_u U + \tau D \mid D=d, X\right)\right] \\
&= E_X\left[\beta_x X + \beta_u E\left(U \mid D=d, X\right) + \tau d\right] \\
&= E_X\left[\beta_x X + \beta_u \underbrace{\left(\frac{d-\alpha_x X}{\alpha_u}\right)}_{\text{From the treatment equation}} + \tau d\right] \\
&= \left(\tau + \frac{\beta_u}{\alpha_u}\right) d + \left(\beta_x - \frac{\beta_u \alpha_x}{\alpha_u}\right) E\left(X\right)
\end{aligned}
$$

Therefore, we have the following,

$$
E_X\left[E\left(Y \mid D=1, X\right) - E\left(Y \mid D=0, X\right)\right] = \left(\tau + \frac{\beta_u}{\alpha_u}\right)(1-0) = \tau + \frac{\beta_u}{\alpha_u}
$$

Sensitivity analysis: a toy model

With a closed-form solution for the bias, how do we use it?

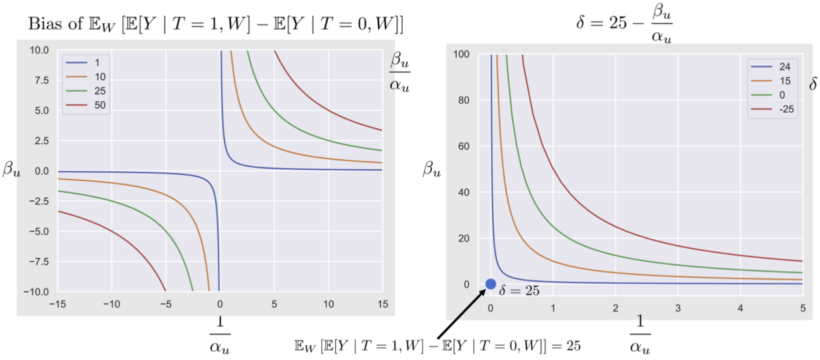
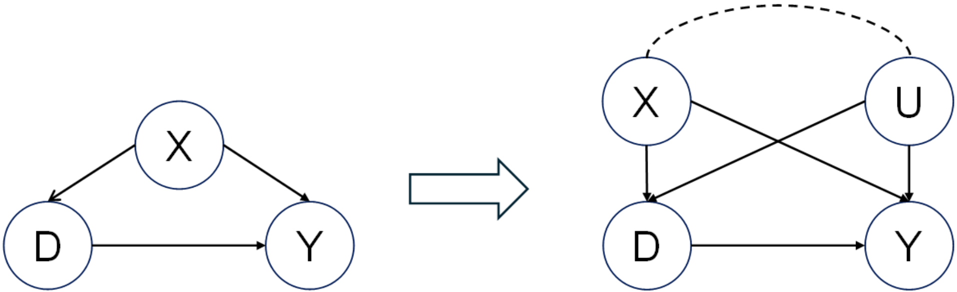


Figure 4: Contour Plots of Bias (left) and "True" Treatment Effect (right)

A unifying framework of sensitivity analysis

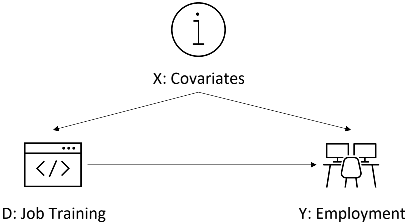


Different sensitivity analyses differ in how U is simulated.

Sensitivity analysis: Rosenbaum's approach for matching

Rosenbaum (1987) developed a sensitivity analysis for matching.

- How **a job training program** (D) can help the unemployed **to find a job** (Y).



Sensitivity analysis: Rosenbaum's approach for matching

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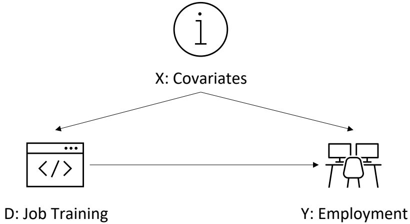
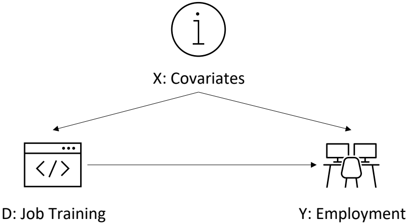


Sensitivity analysis: Rosenbaum's approach for matching

Rosenbaum (1987) developed a sensitivity analysis for matching.

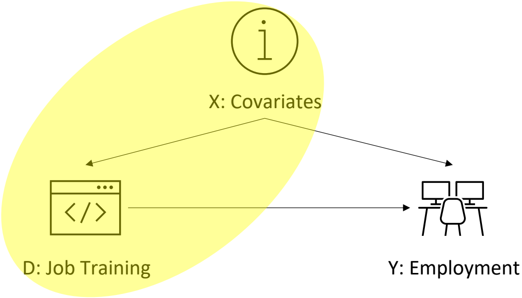
- How **a job training program** (D) can help the unemployed **to find a job** (Y).
- The participation is self-selected, but covariates X are observed.
- Define the propensity score $\pi_i = P(D_i = 1 \mid X_i)$.



Sensitivity analysis: Rosenbaum's approach for matching

Rosenbaum (1987) developed a sensitivity analysis for matching.

The confounder U is simulated based on $P(D_i|X_i)$.



Sensitivity analysis: Rosenbaum's approach

Observation

For **a matched pair** with the same covariates X_i (e.g., age, gender...), the propensity score should be equal, if there is no confounders.

$$\begin{cases} \text{Odds}_{\text{Bob}} &= \dfrac{\pi_{\text{Bob}}}{1-\pi_{\text{Bob}}} \\ \text{Odds}_{\text{Jim}} &= \dfrac{\pi_{\text{Jim}}}{1-\pi_{\text{Jim}}} \end{cases} \text{ and } \Gamma = \dfrac{\dfrac{\pi_{\text{Bob}}}{1-\pi_{\text{Bob}}}}{\dfrac{\pi_{\text{Jim}}}{1-\pi_{\text{Jim}}}} = 1$$

Sensitivity analysis: Rosenbaum's approach

Existence of confounders

If no confounders exist, $\Gamma = 1$. The existence of confounders $\rightarrow \Gamma \neq 1$.

$$\Gamma = \frac{\dfrac{\pi_{\text{Bob}}}{1-\pi_{\text{Bob}}}}{\dfrac{\pi_{\text{Jim}}}{1-\pi_{\text{Jim}}}} \neq 1$$

- Γ represent the severity of the confoundedness.
- Increasing the value of Γ to see the changes in the treatment effects.

Sensitivity analysis: Rosenbaum's approach

		Job Training	
		$D=0$	$D=1$
Employment	$Y=0$	10	c $=50$
	$Y=1$	b $=20$	140

- Suppose use McNemar's exact p-value to test the difference in the employment rate with vs. without the job training ($n = b + c$).

$$\text{p-value} = 2\sum_{k=b}^{n} \binom{n}{k} \left(\frac{\Gamma}{1+\Gamma}\right)^{k} \left(\frac{1}{1+\Gamma}\right)^{n-k}$$

- Under unconfoundedness, $\Gamma = 1$. With a larger Γ, more serious confounding.

Sensitivity analysis: Rosenbaum's approach

p-value	prob $\left(\frac{\Gamma}{1+\Gamma}\right)$	Γ
0.004	0.5000	1.00
0.0018	0.5238	1.10
0.0057	0.5455	1.20
0.0148	0.5652	1.30
0.0328	0.5833	1.40
0.0635	0.6000	1.50

- When $\Gamma = 1.5$, the p-value is larger than 0.05.
- To nullify the effect: a confounder that makes Bob 1.5 times more likely to join the program than Jim.
- Is $1.5\times$ a good indicator of validity? You be the judge!

Sensitivity analysis: Rosenbaum's approach

Technical proof: The probability in McNemar's exact p-value is equal to $\Gamma/(1+\Gamma)$.

The probability in the McNemar's test is θ_i, indicating $P\left(d_i = 1 \mid d_i + d_j = 1\right)$, or the probability of a subject i is treated, conditional on one of the matched pairs $\langle i, j \rangle$ is treated.

$$\begin{aligned}
\theta_i &= \frac{P\left(d_i = 1, d_j = 0\right)}{P\left(d_i = 1, d_j = 0\right) + P\left(d_i = 0, d_j = 1\right)} \\
&= \frac{\pi_i\left(1 - \pi_j\right)}{\pi_i\left(1 - \pi_j\right) + \pi_j\left(1 - \pi_i\right)}
\end{aligned}$$

Then we must have

$$\frac{\theta_i}{1 - \theta_i} = \frac{\pi_i\left(1 - \pi_j\right)}{\pi_j\left(1 - \pi_i\right)} \overset{\text{Definition}}{=} \Gamma$$

Therefore, $\theta_i = \Gamma/(1+\Gamma)$.

Sensitivity analysis: selection bias

- Rosenbaum's approach focuses primarily on matching.
- In practice, we often use a linear regression: $Y = \beta X + \tau D + \varepsilon$.
- Oster 2019 developed a general method based on Altonji et al. 2005.

Variables	**Model 1**	**Model 2**	**...**	**Full Model**
Treatment D	1.00	1.20	...	1.50
X_1	Excl.	Incl.	...	Incl.
...	...	...	...	...
X_K	...	...	...	Incl.

Table 1: Stability of Treatment Effects across Models

Sensitivity analysis: selection bias

Is the stability of the treatment effect sufficient?

Variables	Null Model	Full Model
Treatment D	1.49	1.50
X_1	Excl.	Incl.
...	...	...
X_K	Excl.	Incl.

Two alternative explanations for stability

1. The bias from confounders are minimal.
2. Control variables are not "controlling anything."

Sensitivity analysis: general selection bias

To examine the "efficacy" of the control variables, we need to check the model fit.

Variables	Null Model	Full Model
D	1.49	1.50
X_1	Excl.	Incl.
...	...	...
X_K	Excl.	Incl.
R^2	10.00%	10.01%

Bad Controls

Variables	Null Model	Full Model
D	1.49	1.50
X_1	Excl.	Incl.
...	...	...
X_K	Excl.	Incl.
R^2	10.00%	90.01%

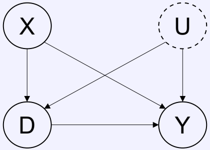
Good Controls

Sensitivity analysis: selection bias

- Observed variables could inform the selection on unobservables.
- We need to consider the **changes in model fit** (e.g., R^2), along with the **changes in coefficients**.

The setup of Oster's sensitivity analysis

$$Y = \alpha + \beta D + \varphi X + \omega U + \varepsilon$$



Both X and U can include multiple variables.

Sensitivity analysis: selection bias

Assumptions of Osters' sensitivity analysis

The degree of selection on unobservables can be approximated by the degree of selection on observables using the model fit changes.

$$\underbrace{(\beta_1 - \beta^*)}_{\text{Selection on Unobs.}} = \boldsymbol{\delta} \underbrace{(\beta_0 - \beta_1)}_{\text{Selection on Obs.}} \frac{R^2_{max} - R^2_1}{R^2_1 - R^2_0}$$

- δ is the sensitivity parameter, indicating the relative importance of unobs. over the observables.
- β_0 and R^2_0 are the treatment effect and R-squared of the uncontrolled model $Y \sim D$.
- β_1 and R^2_1 are the treatment effect and R-squared of the controlled model $Y \sim D + X$.

Sensitivity analysis: selection bias

To understand the result, we do a little bit algebra and assume $\delta = 1$.

$$\beta^* = \beta_1 - (\beta_0 - \beta_1)\frac{R^2_{max} - R^2_1}{R^2_1 - R^2_0}$$
$$\Longrightarrow \frac{\beta_1 - \beta^*}{\beta_0 - \beta_1} = \frac{R^2_{max} - R^2_1}{R^2_1 - R^2_0}$$

- Under the equal selection assumption, the ratio of the movement in coefficients is equal to the ratio of the movement in R-squared.
- The changes in model fit are informative for the changes in coefficients!

Sensitivity analysis: selection bias

To apply the result, we can follow these steps:

1. Run a regression of $Y \sim D$ and record β_0 and R_0^2;

Sensitivity analysis: selection bias

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2. Run a regression of $Y \sim D + X$ and record β_1 and R_1^2;

Sensitivity analysis: selection bias

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1. Run a regression of $Y \sim D$ and record β_0 and R_0^2;
2. Run a regression of $Y \sim D + X$ and record β_1 and R_1^2;
3. Examine the value of δ that nullify the effects or $\beta^* = 0$.

Sensitivity analysis: selection bias

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1. Run a regression of $Y \sim D$ and record β_0 and R_0^2;
2. Run a regression of $Y \sim D + X$ and record β_1 and R_1^2;
3. Examine the value of δ that nullify the effects or $\beta^* = 0$.
4. Discuss whether the value of δ is reasonable.

Sensitivity analysis: challenges

- **Subjective parameter benchmarks**: Hard to interpret sensitivity parameters without clear, objective guidelines

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- **Modeling unobserved confounders**: Simulating hidden confounders relies on strong assumptions.

Sensitivity analysis: challenges

- **Subjective parameter benchmarks**: Hard to interpret sensitivity parameters without clear, objective guidelines
- **Modeling unobserved confounders**: Simulating hidden confounders relies on strong assumptions.
- **Mixing power with identification**: misleading sensitivity analysis from small or large samples.

1. Understanding assumptions in causal inference

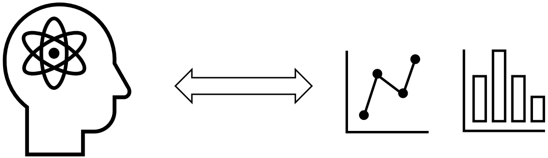
2. Assessing assumptions
   - Sensitivity analysis
   - Consistency tests

3. Conclusions

Consistency tests

Definition (Consistency tests)

If unconfoundedness is a valid assumption, we should observe data patterns that are consistent with it.

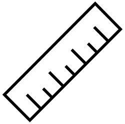


Theoretical Predictions   Data Patterns

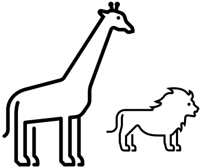
Consistency tests

Identification assumptions and data as a measurement tool for causal effects.

A Measurement Tool



Identification:
Assumptions + Data



Prior Knowledge:
Giraffe is taller!

Measure of Giraffe
> Measure of Lion

Three types of consistency tests

Using pseudo-outcomes, pseudo-treatments, or alternative groups where researchers have prior knowledge of the treatment effect.

Pseudo Outcomes

Pseudo Treatment

Alternative Groups

Consistency tests: pseudo-treatment

Pseudo-outcomes (Dube et al. 2013)

Objective: Spillover effects of US gun laws on gun-related crimes in Mexico.

Population: Mexican municipalities close to the U.S. border.

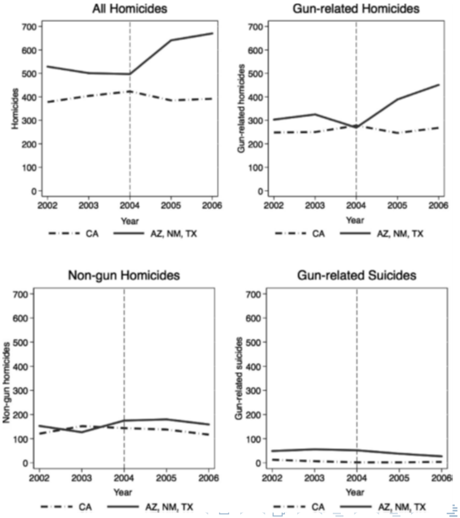
Treatment: Assault weapons from neighboring US state.

Outcome: Gun-related homicides.

Pseudo-outcomes: Accidents, non-gun homicides and suicides.

Consistency tests: pseduo-treatment

Pseudo-outcomes (Dube et al. 2013)



Consistency tests: psuedo-treatment

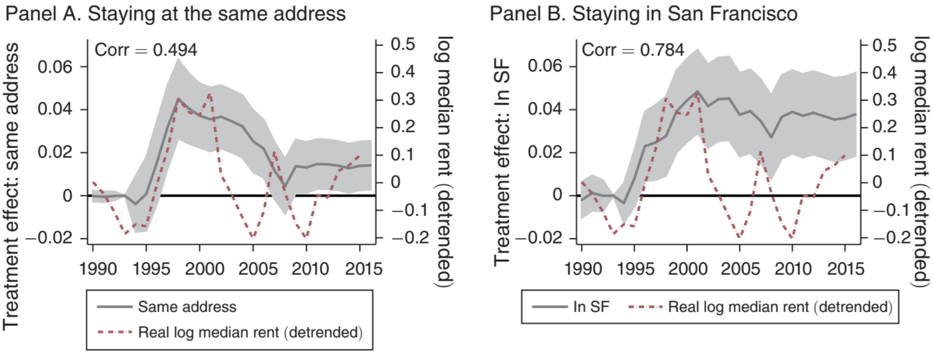
Pseudo-treatment

- In DID, falsification tests that assume the treatment happens in the pre-treatment periods.
- Expecting no effect of these pseudo-treatments.

- In RDD, falsification tests that assume different cutoffs of the forcing variables.
- Expecting no effect of the pseudo-treatments from these cutoffs.

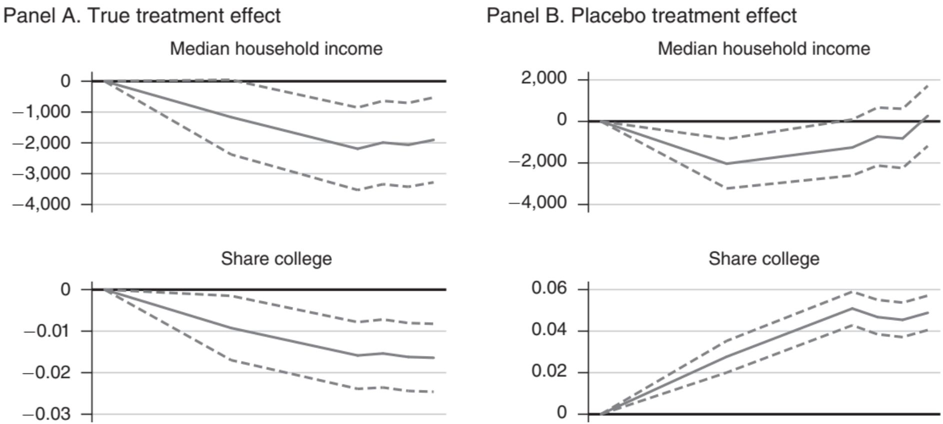
Consitency tests: alternative group

Diamond et al. (2019) compare rent-controlled **buildings before 1980** (treated) to similar **buildings after 1980** (control) to estimate the impact of rent control (D) on tenant mobility and housing supply (Y).



Consistency tests: alternative group

People who move away are used as a “pseduo” treatment group - no treatment effect expected.



What do consistency tests actually test?

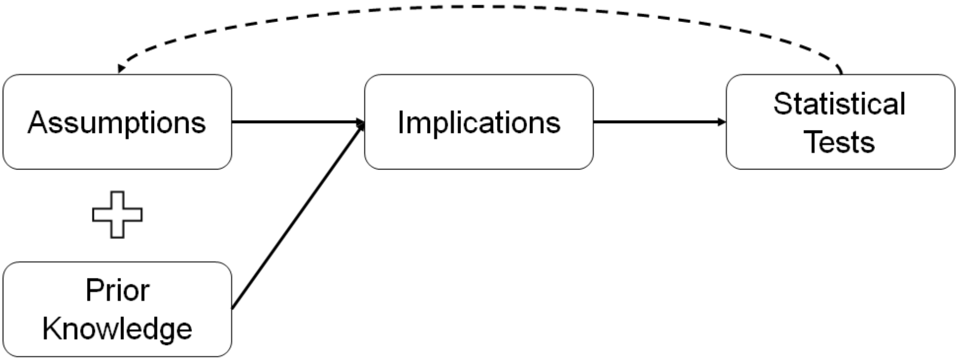
Compare two consistency tests in a matching design:

Tests	**Results**
Covariates balance test	No imbalance
Pseudo pre-treatment	No treatment effect

Question: **why the testing results lend us more confidence in the fundamental assumptions?**

What do consistency tests actually test?

How do test results inform us of the credibility of assumptions?



Statistical testing as plausibility reasoning

Logical resoning

$$\frac{\begin{array}{c}\text{If } A \text{ is true, then } B \text{ is true.}\\ A \text{ is true.}\end{array}}{\text{Conclusion: } B \text{ is true.}}$$

Plausibility reasoning

$$\frac{\begin{array}{c}\text{If } A \text{ is true, then } B \text{ is true.}\\ B \text{ is true (false).}\end{array}}{\text{Conclusion: } A \text{ becomes more (less) plausible.}}$$

Statistical testing as plausibility reasoning

- A = The person is the burglar.
- B = The broken window, the mask and the loot.

$$\frac{\text{If } A \text{ is true, then } B \text{ is true.} \quad B \text{ is true.}}{A \text{ becomes more plausible.}}$$

Consitency tests as plausibility reasoning

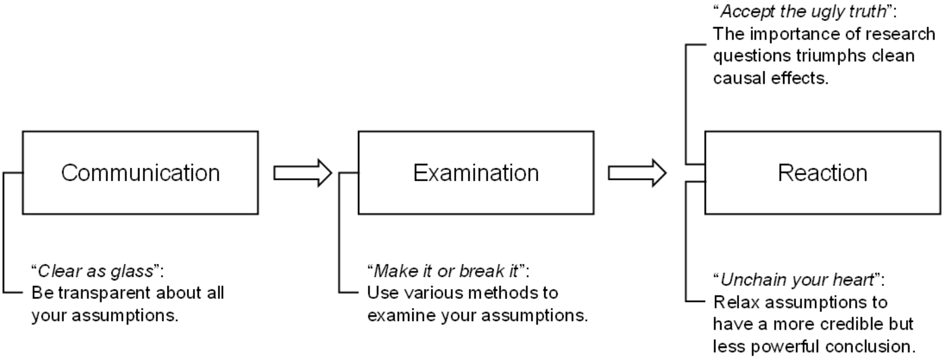
Consistency tests is an application of plausiblity reasoning:

Test	A	B
Covariate Balance	Unconfoundedness	No Imbalance
Pseudo pre-treatment	Unconfoundedness	No Treatment Effect

$$\frac{\begin{array}{c}\text{If } A \text{ is true, then } B \text{ is true.}\\ B \text{ is true.}\end{array}}{A \text{ becomes more plausible.}}$$

1. Understanding assumptions in causal inference
2. Assessing assumptions
3. Conclusions

Summary: a framework for assumptions



Summary: for your own projects

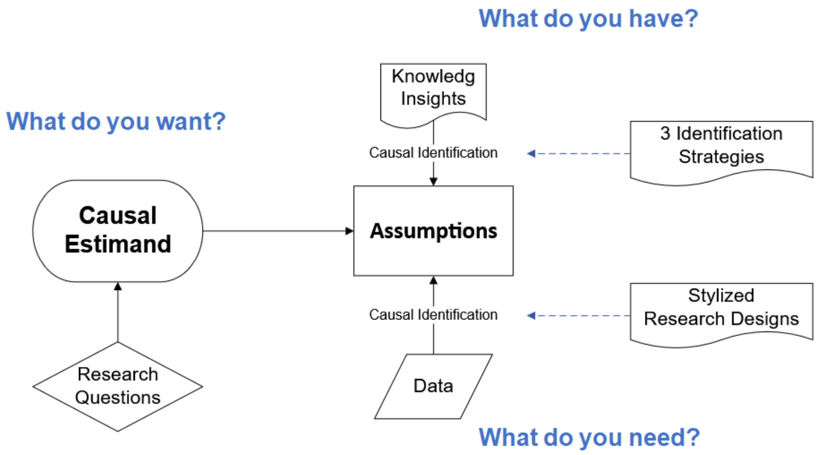
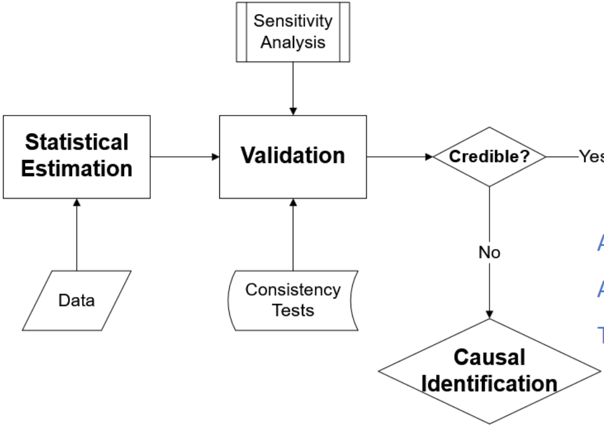


Figure 5: The Identification Stage

Summary: for your own projects



Assumptions Are Not Unique

Assumptions + Data = Conclusions

The Law of Decreasing Credibility

Figure 6: The Estimation Stage

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